```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

// LCD Setup (I2C Address — change if scanner shows different)

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Ultrasonic Sensor Pins

#define trig1 22

#define echo1 23

#define trig2 24

#define echo2 25

#define trig3 26

#define echo3 27

#define trig4 28

#define echo4 29

// Traffic Signal Pins (R, Y, G)

#define R1 30

#define Y1 31

#define G1 32

#define R2 33

#define Y2 34

#define G2 35

#define R3 36

#define Y3 37

#define G3 38

#define R4 39

#define Y4 40

#define G4 41

// Function to calculate distance

Long getDistance(int trigPin, int echoPin) {
```

```
digitalWrite(trigPin, LOW);

delayMicroseconds(2);

digitalWrite(trigPin, HIGH);

delayMicroseconds(10);

digitalWrite(trigPin, LOW);

long duration = pulseIn(echoPin, HIGH, 25000); // Timeout after 25ms

long distance = duration * 0.034 / 2;

if (distance == 0 || distance > 400) distance = 400; // Avoid invalid readings

return distance;

}

// Function to determine green light time based on distance

Int getGreenTime(long distance) {

If (distance <= 10) return 10;   // Very High Density

Else if (distance <= 20) return 8; // High

Else if (distance <= 35) return 6; // Medium

Else return 4;                 // Low

}

// Function to control each lane

Void controlLane(int lane, int R, int Y, int G, int greenTime) {

Serial.print("\n--- Lane ");

Serial.print(lane);

Serial.println(" ---");

Serial.print("Green ON for ");

Serial.print(greenTime);

Serial.println(" sec");

// Green ON

digitalWrite(R, LOW);

digitalWrite(G, HIGH);
```

```
digitalWrite(Y, LOW);
delay(greenTime * 1000);
// Yellow ON
digitalWrite(G, LOW);
digitalWrite(Y, HIGH);
Serial.println("Yellow ON (2 sec)");
Delay(2000);
// Red ON
digitalWrite(Y, LOW);
digitalWrite(R, HIGH);
Serial.println("Red ON");
}
Void setup() {
Serial.begin(9600);
Lcd.begin(16, 2);
Lcd.backlight();
// Sensor pins
pinMode(trig1, OUTPUT); pinMode(echo1, INPUT);
pinMode(trig2, OUTPUT); pinMode(echo2, INPUT);
pinMode(trig3, OUTPUT); pinMode(echo3, INPUT);
pinMode(trig4, OUTPUT); pinMode(echo4, INPUT);
// Signal pins
Int signals[] = {R1, Y1, G1, R2, Y2, G2, R3, Y3, G3, R4, Y4, G4};
For (int i = 0; i < 12; i++) pinMode(signals[i], OUTPUT);
// Initially all red
digitalWrite(R1, HIGH);
digitalWrite(R2, HIGH);
digitalWrite(R3, HIGH);
```

```
digitalWrite(R4, HIGH);

lcd.clear();

lcd.print(“Smart Traffic Sys”);

Serial.println(“System Started…”);

Delay(2000);

}

Void loop() {

// Read all distances

Long d1 = getDistance(trig1, echo1);

Long d2 = getDistance(trig2, echo2);

Long d3 = getDistance(trig3, echo3);

Long d4 = getDistance(trig4, echo4);

// Calculate green times

Int t1 = getGreenTime(d1);

Int t2 = getGreenTime(d2);

Int t3 = getGreenTime(d3);

Int t4 = getGreenTime(d4);

// Print to Serial Monitor

Serial.println(“\n================================”);

Serial.println(“  Density Based Traffic System”);

Serial.println(“================================”);

Serial.print(“Lane 1: “); Serial.print(d1); Serial.print(“ cm  -> Green: “); Serial.print(t1); Serial.println(“s”);

Serial.print(“Lane 2: “); Serial.print(d2); Serial.print(“ cm  -> Green: “); Serial.print(t2); Serial.println(“s”);

Serial.print(“Lane 3: “); Serial.print(d3); Serial.print(“ cm  -> Green: “); Serial.print(t3); Serial.println(“s”);

Serial.print(“Lane 4: “); Serial.print(d4); Serial.print(“ cm  -> Green: “); Serial.print(t4); Serial.println(“s”);
```

```
// Display on LCD (Distance + Time)

Lcd.clear();

Lcd.setCursor(0, 0);

Lcd.print("L1:"); lcd.print(d1); lcd.print("c("); lcd.print(t1); lcd.print("s)");

Lcd.setCursor(0, 1);

Lcd.print("L2:"); lcd.print(d2); lcd.print("c("); lcd.print(t2); lcd.print("s)");

Delay(2000);

Lcd.clear();

Lcd.setCursor(0, 0);

Lcd.print("L3:"); lcd.print(d3); lcd.print("c("); lcd.print(t3); lcd.print("s)");

Lcd.setCursor(0, 1);

Lcd.print("L4:"); lcd.print(d4); lcd.print("c("); lcd.print(t4); lcd.print("s)");

Delay(2000);

// Control Each Lane Sequentially

controlLane(1, R1, Y1, G1, t1);

controlLane(2, R2, Y2, G2, t2);

controlLane(3, R3, Y3, G3, t3);

controlLane(4, R4, Y4, G4, t4);

// Small All-Red Delay before next cycle

digitalWrite(R1, HIGH);

digitalWrite(R2, HIGH);

digitalWrite(R3, HIGH);

digitalWrite(R4, HIGH);

Serial.println("\nAll lanes Red (Safety Delay 3s)");

Delay(3000);

}
```